Route of entry

Route of entry is also important to provide antigenicity and disease. If a microorganism enters through an adverse route, it is degraded by the immune mechanism and is less antigenic. When it enters through its natural route, it will cause disease and it also escapes from the immune system partially. For example: *Vibrio cholerae* entering through the circulatory system will not cause diarrhea; but if it enters through GI tract, it causes diarrhea.

Genotype

The genetic constitution (genotype) of an immunized animal influences the type of immune response the animal manifests, as well as the degree of the response. For instance, human beings are protected against some diseases, yet affected by other diseases. It purely depends upon the genotype.

Dose

To induce antigenicity, an optimum amount of infecting agent or antigen is required and this optimum amount is known as optimum dose. When microbes enter above and below this optimum dose level, fluctuation occurs in antigenicity.

10.5. ANTIBODIES (IMMUNOGLOBULINS)

Antibodies are proteins secretedby B-cells. Antibodies combine chemically with substances which the body recognizes as alien, such as bacteria, virus, and foreign substances in the blood.

Antibodies protect the host against potential parasites by:

1) directly inhibiting binding sites of virus or various enzymes and toxins produced by bacteria,
2) agglutination,
3) opsonization which facilitates removal by phagocytes,
4) lysis of susceptible organisms via complement fixation, and
5) inducing inflammation.

10.5.1. Antibody structure

Antibodies are classified into five types. They are IgG, IgA, IgM, IgD and IgE.

The general structure of antibodies are explained with the example of IgG. The IgG molecule is composed of two identical heavy (H) chains and two identical light (L) chains. The H and L chains are linked by disulfide bridges.

The H chain contains four or five domains. Each domain of immunoglobulin molecule consists of 110 amino acid residues. The amino terminal of heavy chain , shows great sequence variation to other antibodies for different antigens and was therefore called the variable (V) region. The remaining part of the heavy chain is called as constant (C) region. Each of the different heavy chain constant region sequences is called an isotype. The

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